Decision-Making Styles Govern Disfluency Detection in Initial SLA: Challenging Personality-Based Predictions

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While second language acquisition (SLA) research focuses on lexical aspects, non-lexical phenomena like disfluencies are pervasive in spontaneous speech (8% of word count; Zhang, 2020). Although lexically defined in each language (e.g., *um* in English, *euh* in French), these forms are *not formally taught* in instructional or naturalistic settings, requiring learners to recognize and "filter out" disfluencies in streams of foreign sounds on their own. The literature links personality traits (Openness, Conscientiousness, Extraversion, and Agreeableness) to long-term L2 attainment, with Neuroticism viewed as a hindrance (Chen et al., 2022). However, we posit that these predictions derived primarily from interaction and extended learning contexts. Alternatively, the early stage of SLA may draw on different cognitive resources, potentially reversing classic personality patterns. Therefore, we introduce General Decision-Making Style (GDMS; Scott & Bruce, 1995) as a novel predictor of early-stage L2 processing. As early SLA involves rapid hypothesis testing under uncertainty (Birdsong 1994), we hypothesize that analytic styles (Rational and Dependent) may offer an advantage, whereas Intuitive, Avoidant, and Spontaneous styles may hinder discrimination of subtle prosodic cues.

Methods. 210 native English speakers with no prior exposure to French completed a disfluency classification task. We tested their ability to distinguish the French filled pause *euh* from its homophonous lexical counterpart *eux* (both pronounced the non-English vowel /ø/), relying solely on auditory input and without feedback. Participants listened to 24 naturalistic French sentences extracted from corpora beginning with either the filled pause *euh* or the homophonous pronoun (lexical word) *eux* (see Table 1) and determined whether the first word of each utterance was a filler or a lexical word. Stimuli varied in acoustic profile (Fig.1): long *euh* tokens were longer in duration and lower in pitch, whereas short *euh* and *eux* were acoustically similar. After the task, participants provided open-ended comments about their experience and completed the Big Five personality inventory (Goldberg, 1992) and the General Decision-Making Style (GDMS) measure (Scott & Bruce, 1995). Both our raw data and the later converted perceptual sensitivity *d'* values (Hautus et al., 2021) were analyzed using multilevel logistic modeling.

Results. Learners performed well above chance (80%), demonstrating immediate sensitivity to prosodic cues, despite the French vowel /ø/ being absent in native English phonology. Contrasting with traditional SLA predictions, higher Neuroticism (p = .01) and Introversion (p = .046) were associated with higher accuracy (Fig.2A-B), whereas Openness, Conscientiousness, and Agreeableness had no reliable effects (ps < .05). We suggest that in this solitary, analytic task, introverts likely benefited from superior attentional control (Belojevic et al., 2001), while higher Neuroticism may have manifested as beneficial vigilance (Eysenck, 2017), allowing cautious detection of subtle errors without social pressure. Traits linked to long-term motivation (Openness, etc.) showed no effect, suggesting they support sustained effort rather than immediate perceptual discrimination. Decision-making style robustly predicted performance (Fig.2C): Rational (p = .002) and Dependent (p = .004) decision makers significantly outperformed the grand mean, while intuitive learners performed significantly worse than the grand mean (p = .02), rational learners (p = .01), and dependent learners (p = .004). Further analyses showed that dependent decision makers outperformed the average (p = .012) in long *euh* trials, where durational cues were salient, and many reported using a durational rule (Table 2). Rational decision makers performed better than average (p < .001) even under ambiguity (in pronoun *eux* trials), suggesting flexible cue integration as reflected by their comments (Table 3). Neither the avoidant (p = .72) nor spontaneous styles (p = .32) differed significantly from the grand mean.

Discussion. Naïve adult learners successfully acquired disfluency discrimination upon first exposure without feedback, challenging SLA assumptions that acquiring pragmatic and prosodic elements require extended exposure (Pickering, 2004). Our findings suggest that early-stage perceptual sensitivity is governed by domain-general cognitive style and attentional traits, while traditional personality predictors of L2 success exert influence only after extended exposure.

Table 1. Transcript of sample audio stimulus (filler *euh* vs. pronoun *eux*).

Filler *euh*	***Euh**, je monte aussi les moteurs sur certains bateaux.* "Uh, I also install the engines on some boats."
Pronoun *eux*	***Eux**, ils ont une autre religion.* "Them, they have another religion."

Table 2. Sample comments written by a participant of dependent decision-making style.

I found it easy for certain examples because the longer they held the "uhh" sound the more likely it was filler.

Table 3. Sample comments written by a participant of rational decision-making style.

I listened for the tone of the word, whether or not they dragged the first word, and whether or not they seemed to be struggling with saying later parts of the sentence.

Fig.1. Acoustics of the audio stimuli: (A) Duration; (B) F0 (pitch).

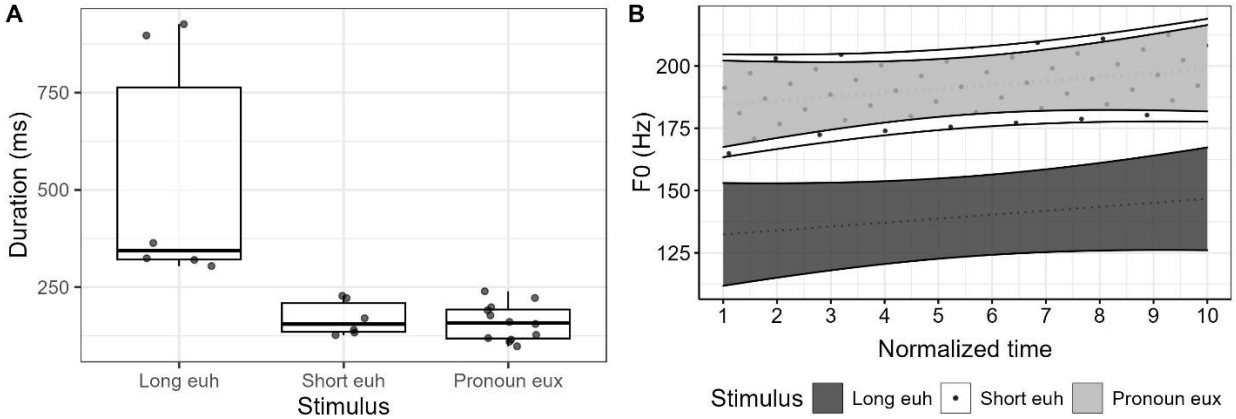
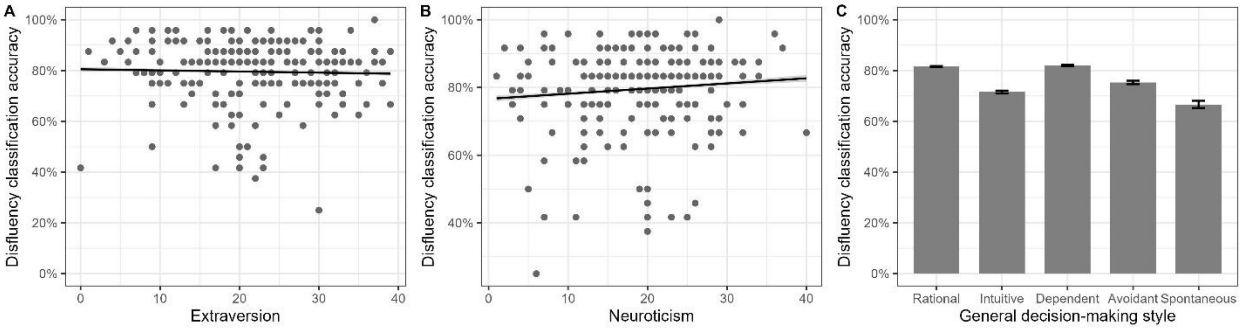
Fig.2. Significant predictors for disfluency classification accuracy: (A) Extraversion; (B) Neuroticism; (C) General decision-making style.